Evaluating a Corpus of Root Causes and Measures to guide RCA processes in Critical Software

Nuno Silva[1,2], Marco Vieira[1]
[1] CISUC, DEI, University of Coimbra, Polo 2 – Universidade de Coimbra, 3030-290 Coimbra, Portugal
[2] CRITICAL Software, SA, Parque Industrial de Taveiro, Lt 48, 3045-504 Coimbra, Portugal
{npsilva, mvieira}@dei.uc.pt

João Cunha[3], Ram Chillarege[4]
[3]Instituto Politécnico de Coimbra – ISEC, DEIS, R. Pedro Nunes, 3030-199 Coimbra, Portugal
[4]Chillarege Inc, 1505 Consett Court, Raleigh, NC 27613-8337, USA
jcunha@isec.pt, ram@chillarege.com

***Abstract*— Reducing defects in critical systems is a challenge faced every day by industry experts and researchers in the dependability and quality assurance fields. However, the state of the art and practice of Root Cause Analysis (RCA) is largely ad-hoc. In this paper we develop a corpus of root causes that are evaluated across a broad survey in the industry and academia. The results provide several key insights including: 1) a broad schema of root causes and actions, leveraging our prior work on real defects in critical systems: 2) understanding on the relevance and value of potential root causes, measures and the RCA process as seen by industry experts across aerospace, safety and academia; 3) detailed insights into the different causes, development measures and their acceptance and importance; 4) survey results providing an invaluable input baseline for further analytical work; and 5) recommendation for an RCA process and data collection schema that could be more widely used and shared across the critical software community.**

Keywords—ODC, critical systems, defect; root cause analysis, dependability, critical systems defects questionnaire;

I. Introduction

Root Cause Analysis (RCA) continues to play a central role in the development of reliable software. This is because software development is a human intensive process that is only as good as the people and the processes that execute it. Improvement comes from learning about the current processes and getting better. That said, the ways in RCA is performed and its effectiveness becomes one of the limiting mechanisms. Over the past 30 years or more, there has been little formalization or sophistication in the mechanisms of RCA as is practiced in the industry at large. Thus, it has remained ad-hoc. Yet, several methods exist in the literature as well as a large number of case studies demonstrating their sophistication and effectiveness (e.g. [4][5][6][13][14][17][18][19][20]). But, industry is still spotty in the use of these methods. Arguably, one of the curses in the software engineering world is the rate of adoption of advanced software engineering methods. These are painfully slow and disorganized across the landscape. Yet, to the man on the street, the perception is the opposite – for what they see it is a rapidly changing industry where obsolescence is built-in. What is known, only to the experts, is that technology changes rapidly, but the methods of process analysis and insight barely register changes across the decades.

In the first part of the paper we review some of the methods supporting RCA, and illustrate them with real world data. Our research work has exploited Orthogonal Defect Classification (ODC), and hereby we summarize one of the studies to illustrate the produced insights. These studies motivated us to develop a survey for the industry to understand the applicability and need of a RCA based process (Research Question RQ1) and the root causes experienced by experts (RQ2). The second part of the paper focuses on the survey and the results we gleaned from it. Following the study, we are able to come full circle, and reflect on where we are in the industry, and what the challenges are to advance the methods and practice of RCA (RQ3), as the one proposed.

There are a few key takeaways that become apparent in the paper, and more that can be generated from a deeper analysis of our data. First: There is a clear necessity to perform detailed RCA in industry. We know this from our experience in working directly with industry through consulting practices. While this is broadly true, it should be less true in the more advanced industry segments (or more regulated) such as aerospace and transportation. Yet, only about half of this select population is aware of more sophisticated RCA methods, and only a quarter are familiar with methods such as ODC. The survey results indicate that most of them (over 75%1) recognize the need for better methods quantification and sophistication. Second: The paper presents a set of root causes, derived from our earlier study [3]. These became the input for the creation of a survey on causes, remedies and process measures. The survey results provide an insight into the current thinking and a basis for future work and discussion. The survey allowed the enhancement of the root causes lists and actions. The results of the survey allow two main novelties: a) the confirmation of the previously defined and applied process through confirmation of the results; b) and improvement of the list of root causes.

This paper contains a short overview of the relevant topics and its state of the art in §II, the summary of the defect analysis process and the main results of the defects analysis up to the main root causes in §III, §IV describes the construction, review and delivery processes of the questionnaire, §V presents the results of the questionnaire and their analysis, and §VI concludes the paper proposing future work.

[1] A conservative number gleaned from data in section VI – aggregating the responses to questions regarding standards, budgets, schedule etc. which only become viable through the use of advanced methods.

1530-2059/17 $31.00 © 2017 IEEE
DOI 10.1109/HASE.2017.19

IEEE computer society

II. BACKGROUND AND RELATED WORK

This section briefly presents preliminary background concepts and relevant related work that supported the process and the root causes and measures identification, namely in what concerns ODC and Independent Software Verification and Validation (ISVV), and the relevant work performed so far that lead to the creation of the questionnaire.

A. Orthogonal Defect Classification (ODC)

ODC [2] is a generic classification technique that turns semantic information into a measurement on the process where defects have been caused, enabling an efficient root cause analysis. ODC is the most commonly used defect classification scheme, but not specifically developed for critical systems. This was also found true for the other classification taxonomies, namely a) Beizer's [23], b) Kaner [24], Falk and Nguyen's , c) Robert Binder's [25], d) Whittaker's "How to Break Software" [26], e) Vijayaraghavan's eCommerce [26], f) Hewlett Packard [27], and g) IEEE Standard Classification for Software Anomalies [28].

The current ODC specification is at version 5.2 [2]. ODC has been adapted for critical systems by the authors in [3].

ODC is used as a starting point for developing new and focused defect taxonomies for particular domains. A few examples of this are Leszak et al. [5] and Lopes Margarido et al. [6], which used ODC as a starting point for studying, building and validating defect categorization schemes.

B. Independent SW Verification and Validation (ISVV)

ISVV [7] is a set of structured engineering activities and tools that allow independent analysts to evaluate the quality of the software engineering artefacts produced at each development phase. These evaluations are performed on mature artefacts, that follow strict engineering standards, and that have previously been verified and validated. ISVV is an additional layer of confidence that should not find severe issues, provided that software has been properly validated.

ISVV produces evidences that support measuring the quality of the software and is referenced in international standards, such as ISO/IEC 12207 [8] and IEEE 1012 [9]. An ISVV guide has been developed for the European Space Agency (ESA) [7]. ISVV is composed by six main phases that can be executed sequentially or adapted as the result of a tailoring process [7]. These phases include (self-explanatory, for more details see [3]): 1) ISVV Planning; 2) Specification/Requirements Verification; 3) Architectural/Design Verification; 4) Source Code Verification; 5) Test Specification/Results Verification; 6) Independent Validation.

C. Related Work

To the best of our knowledge there are no previously published studies performed across industry and academia concerning the opinions of a large group of experts over a defect classification and root cause analysis schema for critical software. We have oriented our analysis towards five different aspects: **a)** a defects classification and root cause analysis structured process; **b)** root causes of defects created during the development; **c)** root causes of failure of defect detection mechanisms; **d)** measures to apply to the development process to avoid/reduce defects; and **e)** verification and validation (V&V) measures to detect and allow correction of the defects. It is worth noting that we are talking about ISVV type of defects, i.e., defects detected after the application of traditional V&V. We present hereby some relevant works related to the aforementioned aspects.

Concerning ISVV defects analysis, some previous studies have been performed with the available datasets, but mostly from the perspective of metrics, efficiency and efficacy of the techniques used to identify the defects and issues ([1] and [11]), and not from point-of-view of the classification or root cause analysis. Furthermore, the research literature does not provide any complete study focused on critical systems defects, nor a complete classification or root cause analysis that relates the results to development lifecycle parameters.

Several researchers have actually analyzed critical systems' failures during different life-cycle phases (from requirements to operations) and performed some empirical studies and root cause analysis ([12], [13] and [14]). For example, Seaman et al. [15] used historical datasets with inspection defect data and applied different categorization schemes for the defects classification. M. Jones [16] has also provided a study about space failures in the frame of the European Space Agency missions, but he concluded that the main cause for all the accidents was only the lack of testing.

For the topic of root cause analysis some relevant works that are applicable to our work are worth mentioning. Neufelder [18] collected data from field defects since 1993 and correlated that data to find the process properties that generated more defects. Rao [19] has made an industry study about root cause defect classification for documentation defects, analyzing only a few dozen defects on a monthly basis. Kumaresh et al [20] conducted a study where a few hundred defects have been classified and the corresponding root causes proposed to the learning of the projects as preventive ideas. None of these works have been performed for critical systems nor with such a complete assessment and coverage of so many defect types, as we did in our work.

Our previous work is described in [3] with process application to a set of 1070 ISVV issues and with an adapted ODC taxonomy for defect type, trigger and impact. The full classification results presented in [3] has been analyzed and the main root causes have been derived and included in the questionnaire [21]. The recommended measures identified from the root causes intend to improve the engineering processes to reduce the issues that are detected or emerge at different stages of the development lifecycle and operations.

III. DEFECTS ANALYSIS PROCESS AND FIELD STUDY

This section presents the defects analysis process developed and applied to an industrial case study and summarizes the results. This process and results have been presented in the questionnaire to the experts in order to cover the research questions highlighted in this work (**RQ1**, **RQ2** and **RQ3**).

A. Defects Analysis Process Summary

As the classification effort and the RCA progressed, we have defined a structured and complete process [3] that is summarized hereafter. This process was presented to the

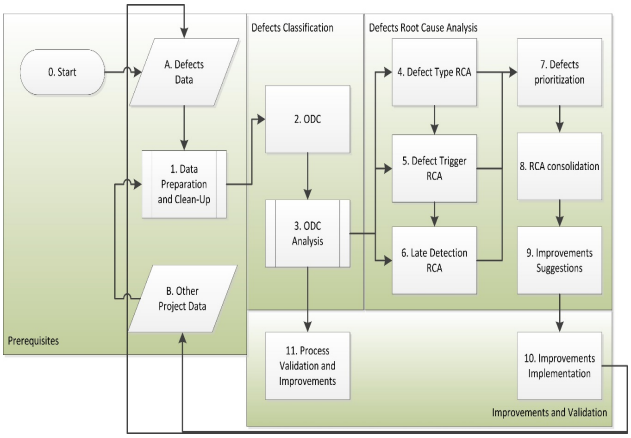

Fig. 1: Defect Assessment Process

respondents of the questionnaire [21], and is composed of four main phases (see Fig. 1):

- **Prerequisites**: Defects data collection and preparation, aggregation of other data if necessary, such as complexity metrics, lifecycle data, etc. A (the issues) and B (the phase of issue introduction, the phase of correction, the type of project, etc.) represent the main process inputs.
- **Defects Classification**: Classification of individual defects according to ODC in order to identify the defect types, triggers and impacts. Note that ODC can be adapted for specific domain and technology purposes.
- **Defects Root Cause Analysis**: Based on the three perspectives (defect type, trigger and impact) identify the root cause of the defect groups (e.g. per type, per trigger). Steps 4 to 6 might be "optional", i.e. it is possible to apply one, two or the three root cause analysis.

TABLE I. ROOT CAUSES FOR MAIN DEFECT TRIGGERS

Root Cause	Defect Trigger
Lack of traceability verification culture	Traceability/Compatibility
Lack or inefficient usage of tools that support traceability across lifecycle phases	
Lack of appropriate test planning and test strategy definition	Test Coverage
Lack or inefficient testing tool and testing environment support	
Incomplete tests specification and execution	
Review process related root causes	Document Consistency/Completeness (Internal Document)
Documentation related root causes	Same as above
Deficient usage of tools and applicable processes	Same as above
Unclear or missing/incomplete specifications	Same as above Logic / Flow
Ambiguous or unclear architecture definition	Logic / Flow
Lack of usage of tools that support data and control flow analysis	Logic / Flow
Inappropriate architecture support tools or tool usage	Design Conformance
Deficient specification or design artefacts	Design Conformance

- **Improvements and Validation**: Act upon the identified root causes, at a process, organizational or resources (human and techniques/tools) level. Measure the effects of the implemented actions. Step 10 represents the actual improvements to the systems under analysis (both environment/organization and processes). Step 11 represents improvements to the classification process.

B. *Field Study Summary*

The results of the application of the defects assessment process (Fig. 1) to a dataset of 1070 defects from space are summarized in this subsection in two main parts: 1) the classification results; 2) the root cause analysis outcome.

1) *Adapted ODC Classification Results*

The summary of the adapted ODC classification results is presented in [3]. The analyzed aerospace ISVV defects dataset contained a large type of Documentation defects (48.1%), but also a relevant amount of Functional defects (19.0%). The main defect triggers are Traceability, Tests Coverage and Consistency/Completeness analysis. The impacts are mainly Capability, Maintenance, Reliability.

The issues that have been classified and that yielded the type and trigger distribution summarized above have been analyzed for root causes and some development and V&V measures have been proposed and described hereafter.

2) *Root Cause Analysis Outcome*

The root cause analysis consisted in a structured process based on a fishbone analysis for the most frequent defect types and defect triggers (see [3]). We have first analyzed the resulting defect types and identified root causes related to development issues, then we have identified the main defect issues according to defect triggers which gave us the V&V weaknesses. In a subsequent step we have proposed a dedicated list of measures to tackle both sets of root causes.

The development defect triggers root causes and recommended measures are presented in [3] and summarized in Q14 and Q18 of [21]. As an example, the list of development defect triggers and corresponding root causes is presented in TABLE I and the corresponding recommended measures in TABLE II with the mapping reference to the relevant defect triggers under square brackets. They are presented in questions Q16 and Q20 of the questionnaire ([21]).

IV. QUESTIONNAIRE

This section presents the process for development and distribution of the questionnaire used as a method to validate the analysis process (**RQ1**) and its results (**RQ2** and **RQ3**).

A. *Definition of the process*

The process of validating and collecting the approval of experts concerning very specific (and sometimes sensitive) topics is not an easy task. We had three objectives with our validation questionnaire: a) to confirm that the proposed process is acceptable and usable by industry; b) to confirm the validity of the identified root causes and measures; and c) to obtain feedback and promote improvements to the process and to the lists of root causes and measures, even if those come from different technical domains (this might be useful in future defect analysis for other datasets).

TABLE II. RECOMMENDED V&V MEASURES

Define appropriate test plans and strategies, especially unit and integration tests. The soundness of the test plans and strategies will reflect in the success of the validation; [Test Coverage]
Ensure appropriate (or automated) traceability analysis at every stage of the development lifecycle; [Traceability/Compatibility]
Improve testing completeness, coverage and reviews; [Test Coverage]
Implement non-functional tests (fault detection, fault injection, redundancy, etc.); [Test Coverage]
Apply/develop tools to validate the implementation and design compliance. [Document Consistency/Completeness; Logic/Flow; Design Conformance]

The first activity was to formulate the questions and have the questionnaire validated (section IV.B). Next, we selected the target groups of respondents (section IV.C). A large group of experts has been contacted from academia and industry worldwide, in particular experts from the dependability and critical systems domains. Finally, the questionnaires have been collected and data analyzed (results in section V).

B. Definition and validation of the Questionnaire

Once we have defined the generic objectives and research questions we have formulated the questions. The questionnaire was composed mainly by:

- A short introduction to the topics, a relevant list of acronyms, a note about the anonymity of the answers, instruction on how and when to deliver the questionnaire, and a short presentation of the defects analysis process;
- Three core parts of the questionnaire:
  - A set of General Questions (to quantify the technical domain and experience, the perceived importance of the role of standards, budget, schedule, external assessment and defects analysis);
  - A set of textual questions collecting feedback about the process (positive and negative) and request for additional recommendations; Another question to ask about the level of recommendation of such a process; (to support **RQ1**)
  - Four sets of questions related with: a) defect development root causes and suggestions; (to support **RQ2**); b) defect detection failure root causes and suggestions; (to support **RQ2**); c) defect avoidance measures and suggestions; (to support **RQ3**); d) V&V measures and suggestions; (to support **RQ3**).
- Additional comments to the questionnaire.

Once the draft structure and questions have been formulated in MS Word, and the questionnaire reviewed internally, a group of 4 experts (two from academia and two from industry) have reviewed and commented the questionnaire for validation. The objective was to have a questionnaire that would be answered by some specific experts, under 1.5 hours and without compromising confidentiality. The comments and improvement from those 4 experts have been taken into account in the final version of the questionnaire.

C. Distribution of the Questionnaire

The questionnaire was delivered directly by email to 171 experts in MS Word and PDF formats and the respondents were given one month to answer. The main selection criteria was that the expert should be working in one of the relevant domains of our study: aerospace, automotive, railway, defense, critical systems. Academic partners with background in dependability, fault injection, V&V were also contacted.

V. RESULTS AND DISCUSSION

The questionnaire [21] contains three main parts: a) general questions related with the perception of the respondents regarding the importance of RCA; b) questions related with our process; and c) questions to evaluate the root causes and measures that resulted from the application of the process to our dataset. This section provides the data analysis performed on the responses, allowing us to make some discussion and conclusions, namely regarding our process.

A. Characterization of the Respondents

The current work is based on **36** received questionnaires. Overall we have received filled questionnaires from relevant players which include National Aeronautics and Space Administration (NASA), European Space Agency (ESA), CISCO, Airbus Defence and Space, Bombardier Transportation, amongst others. Questionnaires from North and South America, Europe and Asia have been received, comprising 6 different academic institutions and over 15 companies.

Not all experts answered the full set of questions, or had opinion on every question. From the received questionnaires, 17 were completely filled, 14 had more than 92% of the questions answered, and 5 had between 56% and 76% of completion. Some experts had no opinion concerning either one of the questionnaire areas or some of the proposed root causes or measures. Overall, all questionnaires have been used to extract useful data.

In the analysis of the questionnaires we have simplified the domains to Academia, Transportation, Aerospace and Others. The distribution of respondents that claimed to work on these domains is: Academia – 19; Transportation – 18; Aerospace – 20; Other – 15. Only 7 researchers specified that they have experience in Academia only, however, for this metric we have also considered researchers who claimed at least 5 years of experience in the academic field. Note that several experts have expertise in several domains. These results provide an interesting base for industrial analysis of the importance and extensibility of our work. The average experience per expert was about 6.7 years for the Academic respondents and 16.7 years for the Industrial respondents.

B. Relevance of RCA

TABLE III shows how the experts perceive the importance of the main drivers considered. Most of the experts consider that the usage of standards is extremely important, that budget and schedule restriction are very important, and that external assessment (such as ISVV or certifications) as well as RCA are also extremely important for critical systems. These results confirm that our motivations for acting upon such systems quality is generally shared by the community.

From questions Q8 and Q9 we verified that 27% have already used ODC in the past, and 57% have used some defect analysis techniques. This is quite surprising for such select

TABLE III. GENERAL QUESTIONS SUMMARY

Questions	ER	VR	SR	NR	NO
Q3: Standards Importance	**53%**	33%	11%	0%	3%
Q4: Budget Importance	30%	**53%**	14%	0%	3%
Q5: Schedule Importance	**42%**	**42%**	11%	0%	6%
Q6: External Assessment Importance	**56%**	30%	11%	0%	3%
Q7: RCA Importance	**61%**	36%	3%	0%	0%

ER (Extremely Relevant), VR (Very Relevant), SR (Somehow Relevant), NR (Not Relevant), NO (No Opinion)

population, since only a little more than half has ever used a structured way to study the (recurrent) defects.

C. Feedback on the Defects Assessment Process

This subsection presents the critics, suggestions and recommendations to the process from the experts in what concerns questions Q10, Q11, Q12. A file containing all the answers to these questions has been compiled in an anonymous way and can be accessed online at [22].

When we asked if the experts would recommend such a process, we got 60% of recommendations 33% of possible recommendation, while only 7% would not recommend such a process. (See Fig. 2 which also includes the part of respondents who did not answered this question).

1) Q10: Process Strengths

From the 32 responses (89%) on the process strengths we can highlight the following ones: i) Being a structured approach; ii) Inclusion of root cause analysis; iii) Inclusion of a classification scheme; iv) Provision of improvements and feedback; and v) Relying on high quality of data.

2) Q11: Process Weaknesses

From the 32 responses on the process weaknesses we can summarize the most relevant/frequent: i) Concerns about how to guarantee the quality of defect data; ii) Large amount of steps in the process; iii) Missing feedback to the process; iv) Difficulty in implementing/enforcing such a process.

These comments are all very relevant. First, the process will only work if the defect data is appropriate. For this, we shall relate comment i) with comment iv), as a cultural enforcement must be broader than just the application of the process, but also cover the defect data collection, the quality checks of the data, and so on. The large amount of steps is required to have the process detailed with simple blocks. Furthermore, the process contains self-feedback to the development and V&V processes.

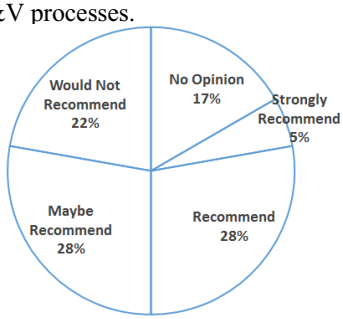

Fig. 2: Process Recommendation Distribution

TABLE IV. AMOUNT OF PROPOSED ROOT CAUSES AND MEASURES

Questions	ER	VR	SR	NR	NO	Total
Q15: Development Defects Root Causes Suggestions	23	26	11	2	46	**62**
Q17: Development Defects Detection Suggestions	15	17	4	0	72	**36**
Q19: Failure of Detecting Defects Root Causes Suggestions	21	10	7	2	68	**40**
Q21: V&V Measures Suggestions	21	13	5	3	66	**42**
TOTAL	**80**	**66**	**27**	**7**	**252**	

ER (Extremely Relevant/Frequent), VR (Very Relevant/Frequent), SR (Somehow Relevant/Frequent), NR (Not Relevant/Frequent), NO (No Opinion) - Questions text simplified to fit the table. Textual description can be found in the questionnaire [27].

3) Q12: Process Suggestions

Only 7 (19%) experts have not provided any additional suggestions to the presented defects assessment process. Overall, out of the three requested suggestions, we had a response rate of 58%.

The summary of **the main and more commonly agreed process suggestions include**: data collection improvements (process, database, quality guarantee); classification/validation activities and data quality check between phases of the process; consideration of projects details/specifics and team dynamics (skills, experience, motivation); and assessment covering also management related issues.

We observe that we have suggestions on the environment and prerequisites, which make absolute sense (data quality, projects details) and also on the validation of the internal process activities, namely the quality of the classification that cannot be easily automated as per today's technologies.

D. Evaluation of the Quality of Root Causes and Measures

The following analysis presents the results and discussion of the experts' opinion on the 4 main results of the application of the defects assessment process to the 1070 ISVV space defects, namely, the development defects root causes, the failure of detecting defects root causes, measures to improve development and V&V measures to improve the defects detection.

The objective of this subsection is to determine if the experts have a common view on the root causes and required measures. Although the presented root causes and measures are naturally associated to the defects dataset and the aerospace domain, we intended to determine if these results are still widely acceptable and probably recurrent also in other domains.

In terms of experts' suggestions, we have analyzed the frequency of words/groups of words and clustered the suggestions of the experts, and those are presented hereby for all the questions where a written opinion was requested.

For the Questions Q15, Q17, Q19 and Q21, the number of provided suggestions (root causes and measures) can be observed in TABLE IV. The experts provided more suggestions to root causes for development problems (62), and a relatively similar amount of Extremely Relevant/Frequent and Very Relevant/Frequent ones (80 and 66).

1) Development Defects Root Causes

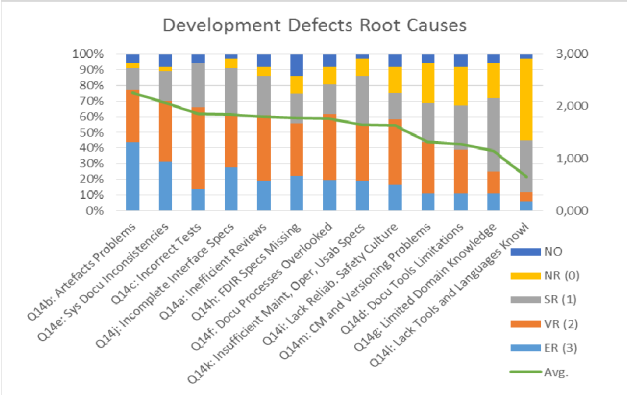

Fig. 3: Development Defects Root Causes

We requested the experts' perception of the frequency of the identified root causes from a development perspective. Q14 presented the main root causes during the development lifecycle. Fig. 3 shows the order of "preference" of the root causes and the relevance level distribution.

No significant change was observed between the overall answers and the answers from experts in Aerospace domain. In what concerns the causes for defects introduced during "development", according to the frequency of occurrence, we have ordered them based on the answers and experts perception (see "Avg." from Fig. 3). Note: This average was calculated providing a weight of 3 to ER, 2 to VR, 1 to SR, and 0 to NR, excluding NO. We can then verify that 9 of the root causes had an Avg. greater than 1.5, meaning that in overall they were evaluated at least as relevant/frequent.

Summary: experts do not consider domain knowledge and tools limitations as relevant/frequent root causes. They consider wrong implementations, incomplete reviews, incomplete tests and specifications for error situations as more frequent causes. This info is important for engineering teams.

Moreover, as a complement we requested the experts to provide some additional root causes from their experience.

The summary/aggregated root causes (Q15) have been collected from the suggestions of 27 experts (75% of the total sample). The **main topics suggested were**: pressures impacting development artefacts (management, schedule/delivery, budget); development/developers overconfidence or carelessness, as well as skills, motivation and interest in the project; systems increasing complexity (technical and team sizes); communication issues (customer, developers, V&V …); and constraints on quality, availability of artefacts, resources skills;

Several experts suggested similar root causes, making the clustering easy. In fact, we observed that some of the suggested root causes are related to our root causes from the questionnaire, e.g. domain knowledge, poor documentation, Fault Detection, Isolation and Recovery (FDIR). Others, as those stated here, are sometimes quite broad but an interesting starting point for consideration in future RCA.

2) Failure of Detecting Defects Root Causes

While analyzing the defects triggers we have identified root causes for failing defect detection. The perception of the frequency of these identified root causes was asked in Q16 and is depicted in Fig. 4.

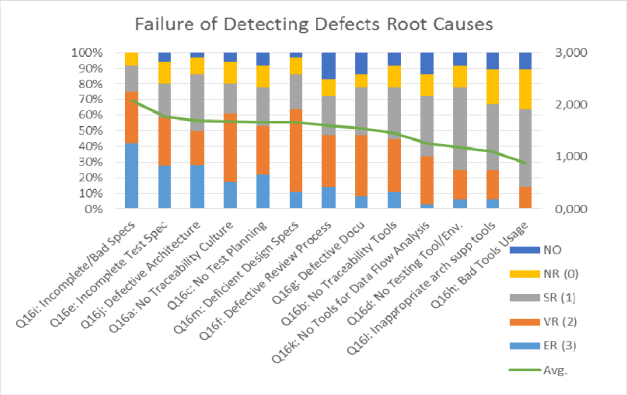

Fig. 4: Failure of Detecting Defects Root Causes

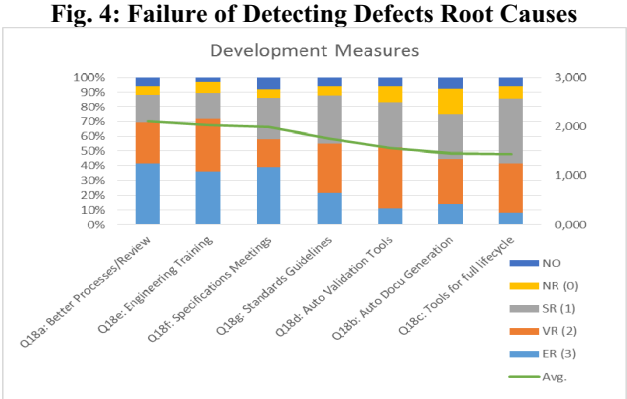

Fig. 5: Development Measures

Besides an improvement in the dominant percentage group of the root causes (excluding Q16j), no significant change has been observed between the overall answers (Fig. 4) and the answers from experts in the Aerospace domain.

Summary: experts do not consider that tools are an important source for not detecting defects. They consider that specifications problems, test specifications and planning and review processes implementation are the main causes. This information is essential for engineering teams.

The summary/aggregated root causes for failure of detecting defects (Q17) have been collected from the suggestions of 18 experts (50% of the total sample). The **main topics suggested were**: knowledge limitations (project, environment, faults, constraints, technologies, …); V&V processes and culture (includes standards); staff inexperience; management related issues (plan, pressures, risks, schedules); and size and complexity of systems (untestable requirements, long testing periods, lack of early involvement of safety/V&V teams). Some of the suggested root causes are already in the questionnaire, e.g. test plan/strategy, or testing tooling/environment issues (also problems with tools).

3) Development Measures

Following the root cause analysis, we have mapped concrete measures to tackle the most frequent root causes and thus avoid the same type of defects. The perception of the relevance of these proposed measures was asked in Q18.

The main observation from Fig. 5 is that Aerospace experts consider that Engineering Training (Q18e) is a measure that could help in reducing the introduction of defects more

than the overall experts do. The existence of specifications meetings to clarify the specifications is also emphasized by the Aerospace experts. Most of the remaining measures are similar to the overall experts' opinion.

Summary: again, experts do not seem to consider that tools are the most important asset for reducing defects during the lifecycle. More importantly, they consider that better processes, clarification meetings with the customer and dedicated engineering trainings are the main solutions proposed.

Additional development measures suggestions have been provided by 20 of the respondents (56% of the total sample) and allowed the **identification of a large set of measures according to the experience of the experts**: require early results, early quality guarantee of requirements (or formal specifications); promote defects knowledge and analysis, fault awareness (FDIR) training, compensations; simplify systems, technologies, management, lifecycles; improve communication, meetings efficiency, milestones with more quality; traceability simplification and implementation; promote teams training and involvement at different phases; and use standards but plan for improvements/additional tasks.

There is a relation between some of the suggestions and measures already defined, namely the enforcement of strict review processes/development processes/documentation processes and the standards' topics. Moreover, some of the suggestions also fit the V&V measures and there are commonalities, such as the traceability topics.

4) V&V Measures

Following the RCA we have performed based on the defect triggers, we mapped concrete measures to tackle the lack of defect detection. The perception of the relevance of these proposed measures was asked in Q20.

Aerospace experts basically consider that Q20c Test Coverage is an important tool (much more than the other experts – depicted in Fig. 6).

Summary: experts still do not seem to consider that tools are the most important asset for V&V to help detecting defects. More importantly, experts suggest that better test plans, improved test coverage criteria and better traceability analysis are the most important V&V actions.

Additional V&V measures suggestions were obtained from 21 of the respondents (58% of the total sample), allowing the identification of a large **set of V&V measures according to the experience of the experts**: more customer involvement in testing / experts involvement for requirements testing; tools to support V&V processes, static analysis, automation of testing tasks; better communication between V&V and development; monitor and measure quality; and enforce early defect detection (model-driven techniques, formal methods, test driven development).

Again, there were commonalities between the proposed measures and some of the measures already included in the questionnaire, for example, the experts also proposed better test plans and strategies, and more abnormal testing (non-functional) - Informal V&V methods. Moreover, some of the V&V measures proposed were already covered as part of the development measures on the questions, for example, better review processes, better code inspections or team training, focus, specialization, defects awareness.

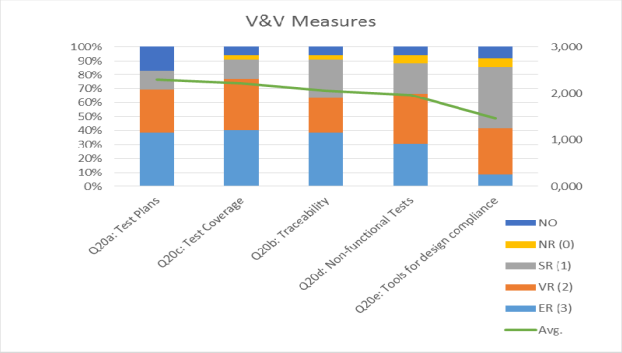

Fig. 6: V&V Measures

E. Discussion

The community (both Academia and Industry) consider that **defect analysis**, **external assessments** and the **use of standards** are of **utmost importance**, in parallel, budget and schedule constraint play an important role in the quality of the projects outcomes. Almost half of the respondents has never performed defect analysis, so this is an area that really needs improvements to be implemented.

The **proposed assessment process** got acceptance from a majority of the respondents. They claimed the process was **structured** and could provide valuable **feedback**; however, they were concerned about how to obtain good **quality of defects data** and how to implement a process with a **large number of steps**. Some of the suggestions were concerning data quality and how to validate the obtained results. We claim that usage over the time and monitoring of the defects frequency and impact would be a good validation method.

The root cause results (development and V&V) have allowed the **ranking of our identified root causes**, and this has shown that the experts consider wrong software implementations, incomplete reviews, incomplete tests or test plans and lack of consideration of error situations to be the most common problem, while lack of domain knowledge and tools (or lack of them) do not seem to be the causes for most defects. This is an interesting results that shows that the critical software domain is still heavily dependent on manual actions, on human skills, and on traditional development (from specifications to tests).

For the **ranking of the measures**, the most voted were review processes, improved communication, test plans & tests coverage, better traceabilities and training. Again, tools and automation are generally the V&V and development assets that seem less important for the experts. This shows once again that the manual and human based techniques are still the ones the community is considering more applicable.

Additional suggestions for root causes and measures have been proposed by the experts. These were very much in-line with those presented in the questionnaire, providing also an interesting addition for future RCA. The proposed root causes and measures arise from specific experience of the experts and provide a good overview of the problems (and possible solutions) for critical systems at large. The conclusion we can make is that our process produced a **very well accepted set of root causes and measures**, and this supports positive-

ly the evaluation of the process and the methods that compose it. With some eventual adjustments we can make it even more widely accepted and usable.

VI. Conclusions and Future Work

The analysis of 1070 ISVV defects by using an RCA process resulted in the identification of development root causes and causes for not detecting the defects, as well as measures suggested to eliminate the defects and to allow their detection during the development lifecycle. This corpus of root causes and measures has been described in a questionnaire delivered to 171 experts and answered by 36 experts, who provided their views on the process and on the soundness of the results. From the answers we conclude that the process is accepted (60.0%) being considered structured and with recurrent feedback. This weakly supports **RQ1**.

In what concerns the root causes of the defects (**RQ2**), it seems that the lack/utilization of tools and the lack of domain knowledge is not considered a main point by the experts. On the contrary, wrong implementations, incomplete reviews, incomplete tests and specifications should be tackled with better processes, more frequent clarification meetings with the customer and dedicated engineering trainings. Moreover, defects detection can be enhanced with better test plans, improved test coverage criteria and better traceability.

Additional root causes and measures have been provided to avoid defects and improve their detection. We have collected/summarized these suggestions that provide some hints for better future root cause analysis. This contributes to **RQ3**.

For future work we intend to tweak the defects assessment process in-line with the experts' suggestions and also: a) finalize a classification and root cause analysis with railway systems defects; b) conclude on the similarities of the results and the applicability of the process (or identify eventual modifications); c) start building a database that connects root causes with measures and make it available for root cause analysis of future datasets (for development and V&V); and d) evaluate how such ODC based assessment processes affect engineering and agile-based development.

Acknowledgments

This work was supported by European Project FP7-2012-324334-CECRIS (CErtification of CRItical Systems) [10].

We thank the 4 experts that provided reviews on the questionnaire and the 36 respondents of the questionnaire.

References

[1] N. Silva, R. Lopes, A. Esper, R. Barbosa, "Results from an independent view on the validation of safety critical space system", DASIA 2013, 14-16 May, 2013, Oporto, Portugal.

[2] Orthogonal Defect Classification v 5.2 for Software Design and Code, IBM, September 12, 2013.

[3] Nuno Silva, João Carlos Cunha, Marco Vieira, A field study on root cause analysis of defects in space software, Reliability Engineering & System Safety, Available online 24 August 2016, ISSN 0951-8320, http://dx.doi.org/10.1016/j.ress.2016.08.016.

[4] R. Chillarege, I. S. Bhandari, J. K. Chaar, M. J. Halliday, D. S. Moebus, B. K. Ray, and M.-Y. Wong, 1992. Orthogonal Defect Classification-A Concept for InProcess Measurements, IEEE Transactions on Software Engineering, vol. 18, pp. 943-956.

[5] M. Leszak, D. E. Perry, and D. Stoll, 2002. Classification and evaluation of defects in a project retrospective, The Journal of Systems and Software, vol. 61, pp. 173-187.

[6] I. Lopes Margarido, J. Pascoal Faria, R. Moreira Vidal, and M. Vieira. "Classification of Defect Types in Requirements Specifications: Literature Review, Proposal and Assessment." In 6th CISTI, 1–6. IEEE, 2011.

[7] ESA ISVV Guide, issue 2.0, 29/12/2008, European Space Agency.

[8] ISO/IEC 12207:2008 Systems and software engineering – Software life cycle processes.

[9] IEEE 1012-2004 - IEEE Standard for Software Verification and Validation. IEEE Computer Society.

[10] CECRIS website, http://www.cecris-project.eu/, visited 17/05/2016.

[11] Silva, N.; Lopes, R., 10 Years of ISVV: What's Next?, Software Reliability Engineering Workshops (ISSREW), 2012 IEEE 23rd International Symposium on , vol., no., pp.361,366, 27-30 Nov. 2012.

[12] Lutz, R.R.; mikulski, I.C., Empirical analysis of safety-critical anomalies during operations, Software Engineering, IEEE Transactions on , vol.30, no.3, pp.172,180, March 2004

[13] M. Leszak, D. E. Perry, and D. Stoll, A Case Study in Root Cause Defect Analysis, Proc 22"d Intl Conf SW Eng (ICSE'OO), IEEE CS Press, Los Alamitos, CA, 2002, pp. 428-437.

[14] K. A. Weiss, N. Leveson, K. Lundqvist, N. Farid, and M. Stringfellow, "An Analysis of Causation in Aerospace Accidents," Space, 2001, Aug., 2001.

[15] Seaman, Carolyn B., Forrest Shull, Myrna Regardie, Denis Elbert, Raimund L. Feldmann, Yuepu Guo, and Sally Godfrey. "Defect Categorization: Making Use of a Decade of Widely Varying Historical Data." In Proceedings of the Second ACM-IEEE International Symposium on Empirical Software Engineering and Measurement, 149–57. ACM, 2008.

[16] Software Engineering: Are we getting better at it?, Michael Jones, ESA Bulletin 121, February 2005, pp. 52-57.

[17] S. Wagner. A Literature Survey of the Quality Economics of Defect-Detection Techniques. In Proc. 5th ACM-IEEE Int. Symposium on Empirical Software Engineering (ISESE '06). ACM Press, 2006.

[18] Neufelder, Ann Marie, "The top ten things that have been proven to impact software reliability", November 30, 2012, http://www.softrel.com/downloads/TopTen.pdf.

[19] Rao, Ramaa, "Root Cause Defect Classification (RCDC) for Documentation Defects", 2014, http://www.stc-india.org/conferences/2014/presentations/Root Cause and Defect Classification for Documentation Bugs - Ramaa Rao.pdf.

[20] Kumaresh, S, Baskaran, R, "Defect Analsysis and Prevention for Software Process Quality Improvement", Int. Journal of Computer Applications (0975 – 8887), Volume 8– No.7, October 2010.

[21] Defects Assessment Questionnaire, https://drive.google.com/file/d/0B1Me2x-y-YiPR01UOVRlUHZOS1E/view?usp=sharing.

[22] Textual answers and recommendations, https://drive.google.com/open?id=0B1Me2x-y-YiPajRqX3RZX1Y1UHc

[23] Beizer B.: Software Testing Techniques. Second Edition, Van Nostrand Reinhold New York (now: Coriolis Press). (1990).

[24] Kaner, C., Falk, J., Nguyen, H.Q.: Testing Computer Software (2nd. Edition). International Thomson Computer Press (1999)

[25] Binder, R.V.: Testing Object-oriented Systems Models, Patterns, and Tools. Addison-Wesley (1999)

[26] Copeland L., Software Defect Taxonomies, http://flylib.com/books/en/2.156.1.108/1/.

[27] Grady, R.B.: Practical Software Metrics For Project Management and Process Improvement. Hewlett-Packard (1992).

[28] IEEE: IEEE 1044-2009 Standard Classification for Software Anomalies. Institute of Electrical and Electronics Engineers (7 Jan 2010).